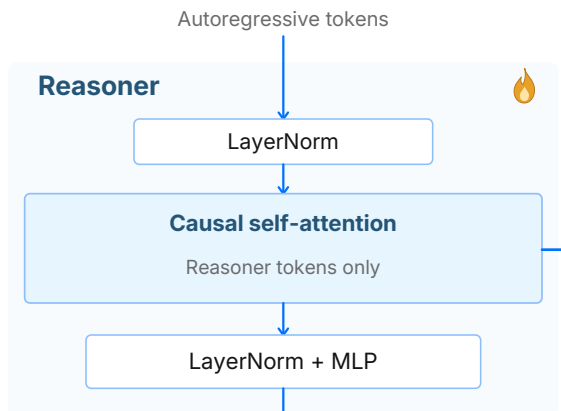
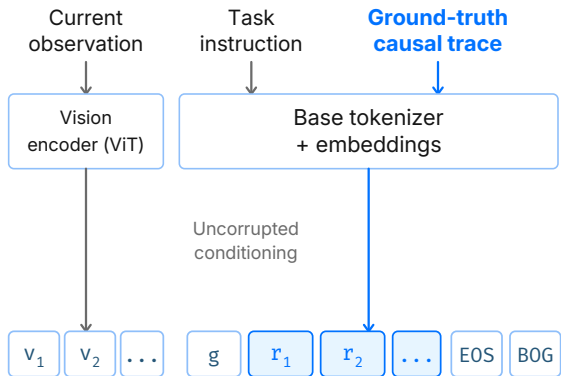


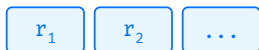


Cosmos3-Nano-Policy + ARC

L

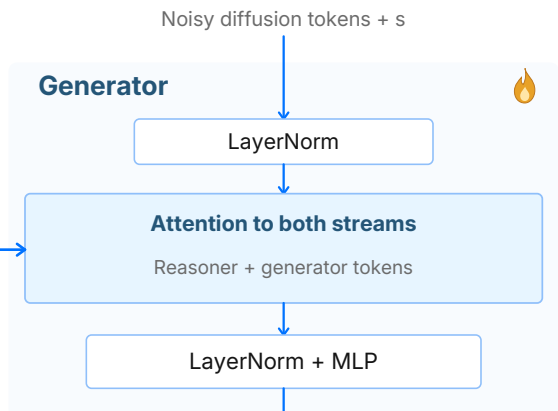
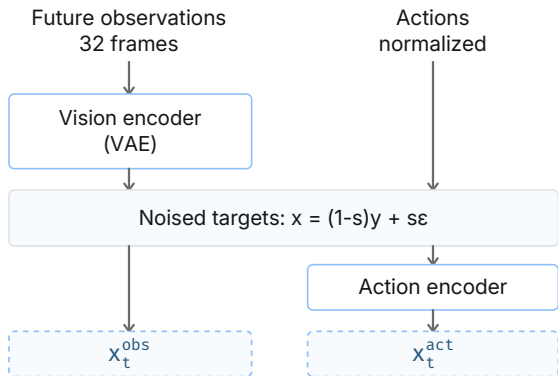


Predicted causal-trace tokens

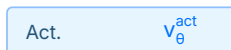
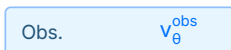


Next-token loss

L_{trace}



Predicted velocities



Observation FM

$$L_{\text{FM}}^{\text{obs}} = E[\text{MSE}_{\text{obs}}(v_\theta^{\text{obs}}, \epsilon^{\text{obs}} - y_t^{\text{obs}})]$$

Action FM

$$L_{\text{FM}}^{\text{act}} = E[\text{MSE}_{\text{act}}(v_\theta^{\text{act}}, \epsilon^{\text{act}} - y_t^{\text{act}})]$$

$$L_{\text{WAM}} = L_{\text{FM}}^{\text{obs}} + 10 L_{\text{FM}}^{\text{act}} + \lambda_{\text{trace}} L_{\text{trace}}$$

Both flow-matching losses also update the reasoner.



Fine-tuned